



UNIVERSITY OF AMSTERDAM
Amsterdam Business School

Data Science in Online Marketing

Media Mix Modeling & Conversion Attribution

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Course Outline

Date	Topic	Assignment
3 april	Recommenders: Content Based	
10 april	Recommenders: Collaborative Filtering	Yes
17 april	Recommenders: Guest Lecture Bol.com Online advertising: Introduction	
24 april	Online advertising: Reinforcement Learning	
1 may	Online advertising: CTR prediction	Yes
15 may	Online advertising: Media Mix Modeling & Conversion Attribution	
22 may	Revision	
29 may	Digital exam on location	

Measuring Performance in online marketing

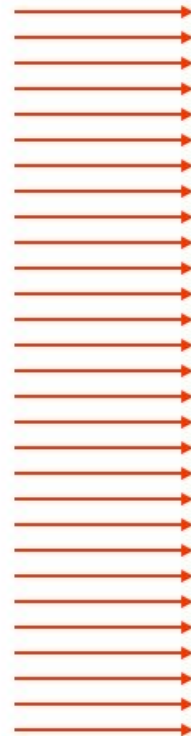
Click-through rate (CTR) =

$$\frac{\# \text{ click throughs}}{\# \text{ impressions}}$$

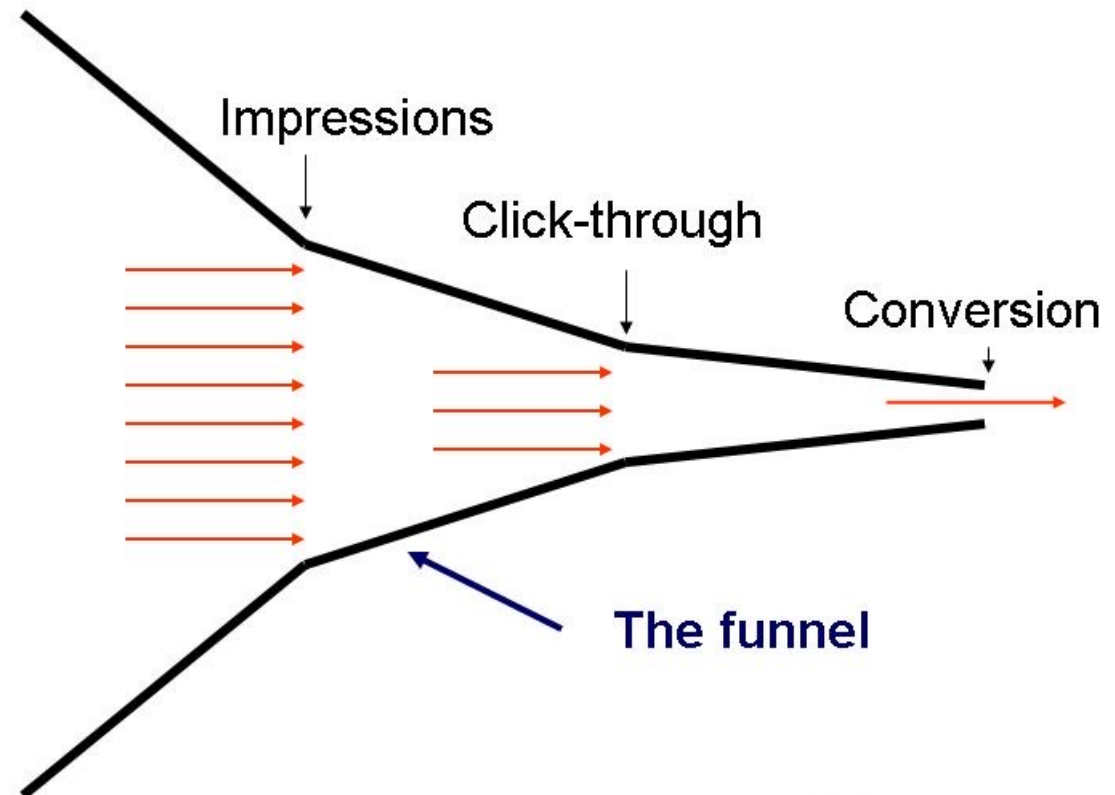
Conversion rate (CR) =

$$\frac{\text{conversions}}{\# \text{ click throughs}}$$

Target audience



The marketing funnel





“Half the money I spend on advertising is wasted; the trouble is I don’t know which half”

Johan Wanamaker

Father of modern advertising and a “pioneer in marketing.”

Online Advertising – Multi Channel

Advertisers want to spend their advertising budget effectively:

- What overall advertising budget do I need?
- How do I allocate it over different channels and campaigns?

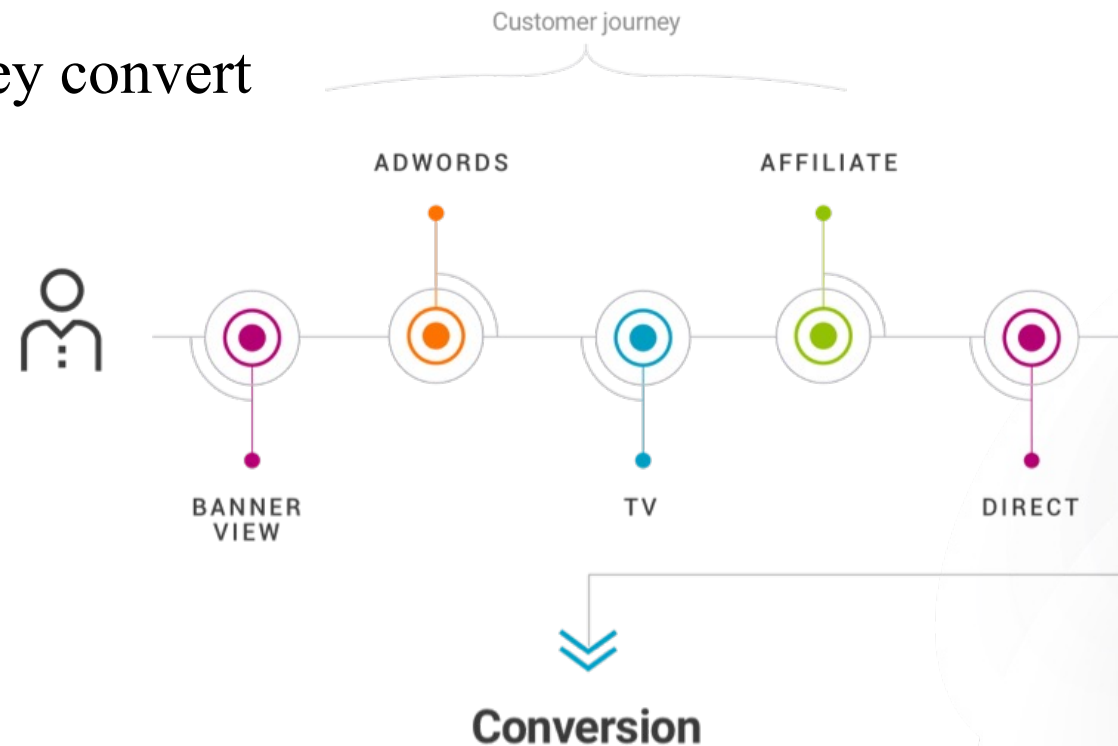
This requires insights into:

- What is the impact of spending more/less on each channel/campaign?
- How do the channels interact?

Online Advertising – Multi Channel

Until now we've look at optimizing individual channels, but ..

- advertisers use multiple channels at the same time
- people see multiple ads before they convert



Multichannel advertising modeling

Media Mix Modeling

Aggregate data on media spend and sales

“We advertised on TV last week and sales went up”

Conversion Attribution

Detailed data on customer touchpoints and conversions

“This person saw my advertisement on nu.nl yesterday and then entered my website. Today he came back and bought something”



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Media Mix Modeling

Media/Marketing Mix Models

Analytical approach that uses historic data to quantify the sales impact of various marketing activities

- Often by use of regression models
- Dependent variable: sales

Understanding this impact is required to take (fact based) decisions on marketing budget levels and allocation

Marketing Mix Model – Linear Regression

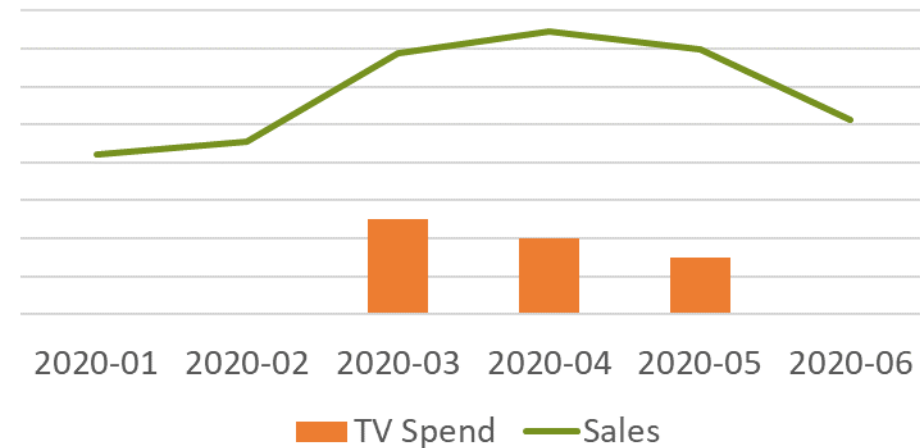
Linear regression model:

$$Q = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k$$

- Q = sales
- X_i = i^{th} explanatory variable

Linear Regression - Example

Week	Sales	TV Spend
2020-01	42,055	
2020-02	45,409	
2020-03	68,792	€ 25,000
2020-04	74,666	€ 20,000
2020-05	69,942	€ 15,000
2020-06	51,111	

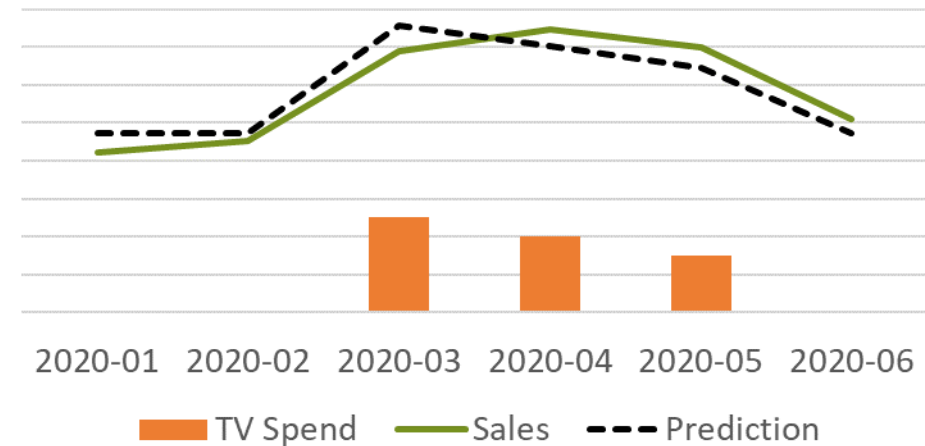


$$Q = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

- Q : sales
- X_1 : TV spend

Linear Regression - Example

Week	Sales	TV Spend
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2020-05	69,942	€ 15,000
2020-06	51,111	



$$Q = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$$

- Base Sales: $\beta_0 = 47,239$
- Incremental sales per euro TV Spend: $\beta_1 = 1.14$

Challenges in Marketing Mix Modelling

- What explanatory variables?
- Multicollinearity between explanatory variables
- What measure for marketing activity
- Diminishing returns
- Lagged effects
- Long term effects (brand equity)
- Reversed causality



Explanatory Variables

Media variables:

- Marketing activities for TV, radio, etc.
- Marketing activities from competitors for TV, radio, etc.
- Interactions between TV, radio, etc. (do they enhance each other?)

Essential to include other variables that impact sales:

- Price
- Promotions
- External effects: seasonality, weather, economy, ..

Multicollinearity

- High risk of multicollinearity between explanatory variables in marketing mix models
- During a campaign, companies often advertise on multiple channels at the same time (TV + radio + billboard)
- But if I always advertise on radio and TV at the same time, then what channel is responsible for the uplift in sales?

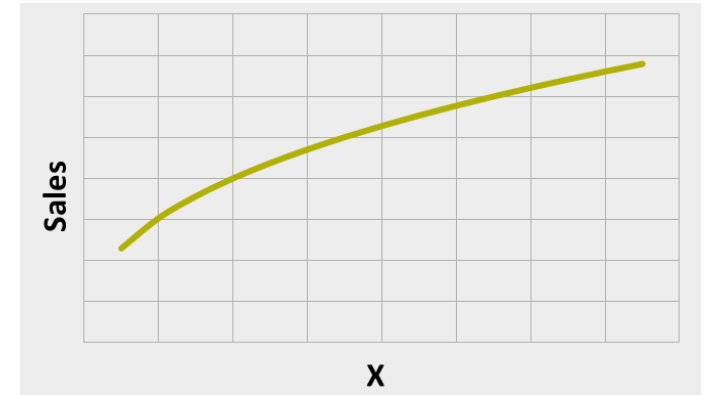
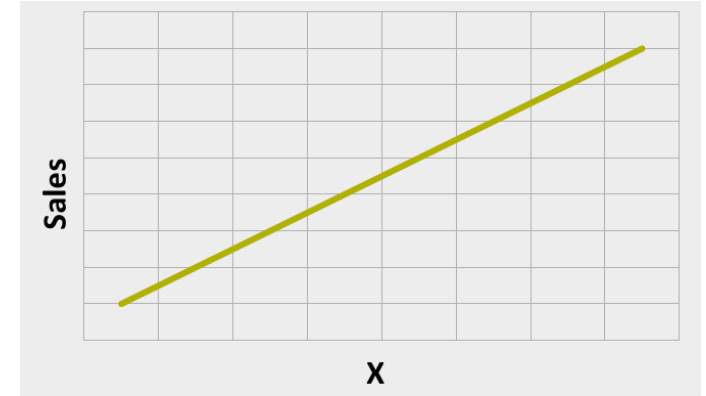
Diminishing Returns

- Linear model does not account for diminishing returns typical for marketing effects

➤ Logarithmic model

- Linear sales model $Q = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k$
- Logarithmic model $Q = \beta_0 + \beta_1 \ln(x_1) + \beta_2 \ln(x_2) + \dots$

Other variable transformations are also possible. Different variables can have different transformations (linear, logarithmic, exponential, ...)

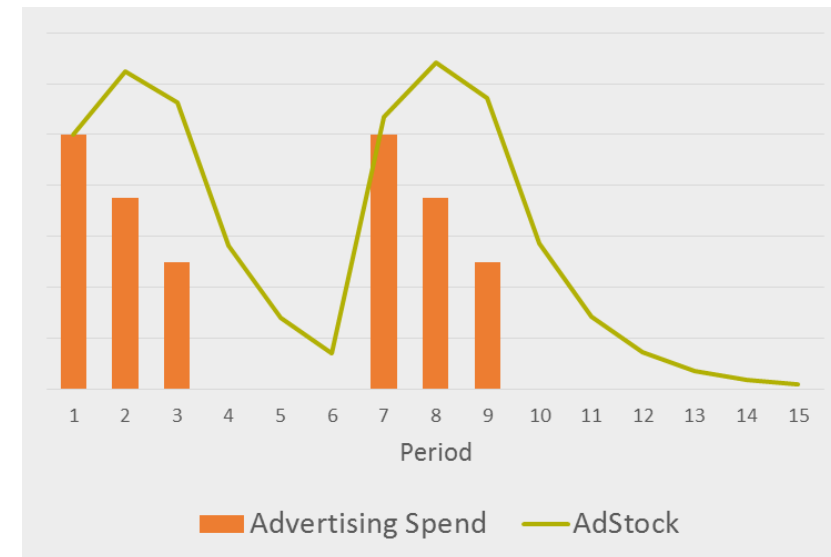


Lagged Effects

- Marketing this week typically also influences sales next week
- Two options:
 - Include lagged variables of the marketing mix variables
 - Include AdStock variable

AdStock

- $A_t = X_t + \lambda A_{t-1}$
- X_t = Marketing activity in period t
- λ = Advertising decay factor, $0 < \lambda < 1$



Lagged Effects

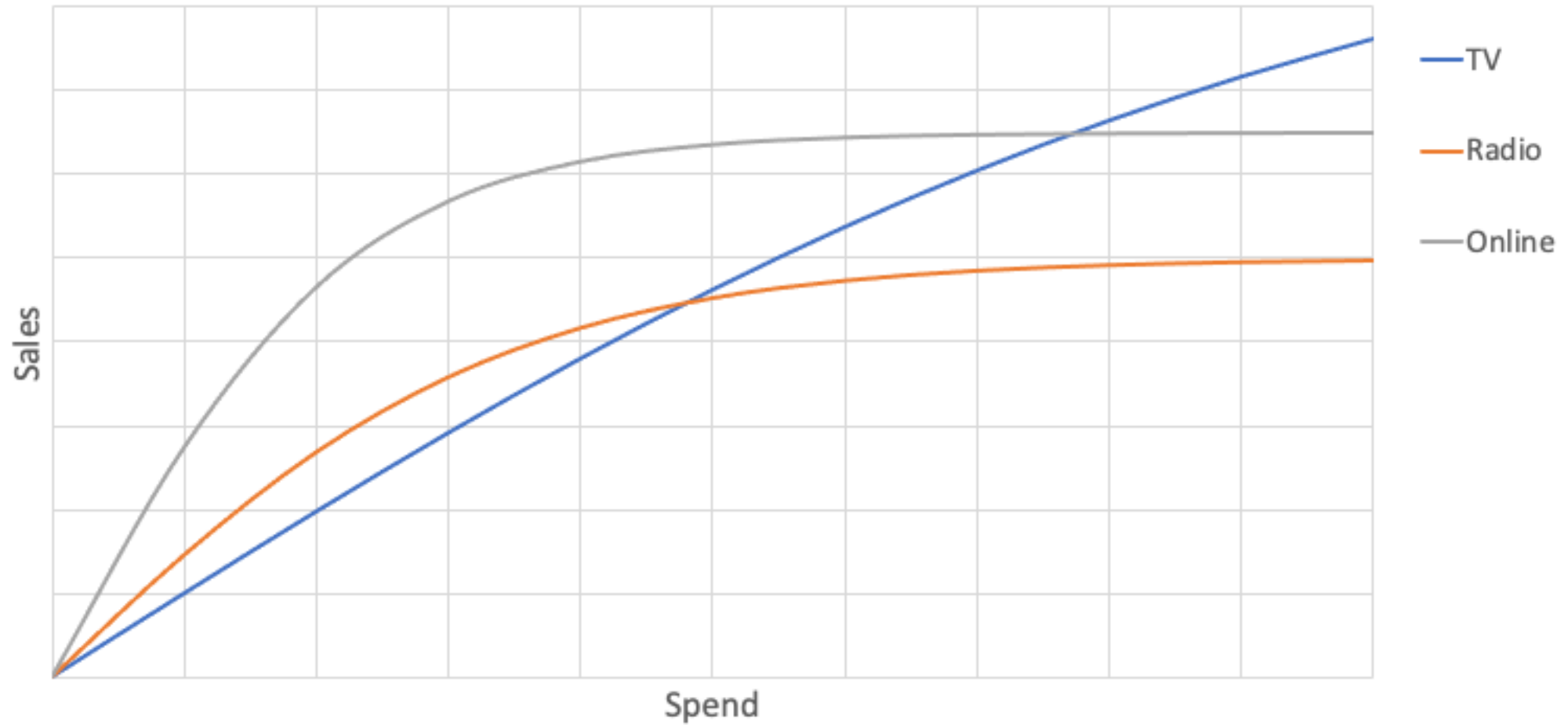
- Disadvantage of the lagged variables is that you can easily get many variables that are often very similar (high risk of multicollinearity)
- Disadvantage of the AdStock variable is that you lose information by capturing multiple lags in one parameter



Media Mix Modelling

- Understand the relation between media spend (and other factors) and sales: **Inference**
- Aggregated data per period
- Common technique: Linear Regression
 - Carefully select variables to include relevant effects, deal with multicollinearity
 - Transformation of variables: AdStock, Logarithms, etc.

Media Mix Modelling – Example output





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Conversion Attribution

Multichannel advertising modeling

Media Mix Modeling

Aggregate data on media spend and sales

“We advertised on TV last week and sales went up”

Conversion Attribution

Detailed data on customer touchpoints and conversions

“This person saw my advertisement on nu.nl yesterday and then entered my website. Today he came back and bought something”

We have all the data, but...

Assume we have 3 ads:

- 2% of the people that click on our CNN ad convert
 - 1.5% of the people that click on our Facebook ad convert
 - 2.2% of the people that click on our Google ad convert
-
- What about people that saw all 3?
 - How do we attribute the conversion over the 3 different channels?



Conversion Attribution

Online we do know which people saw which advertisements during the customer journey and if they converted.

- Use detailed data about customer journeys to estimate effectiveness of channels and ads

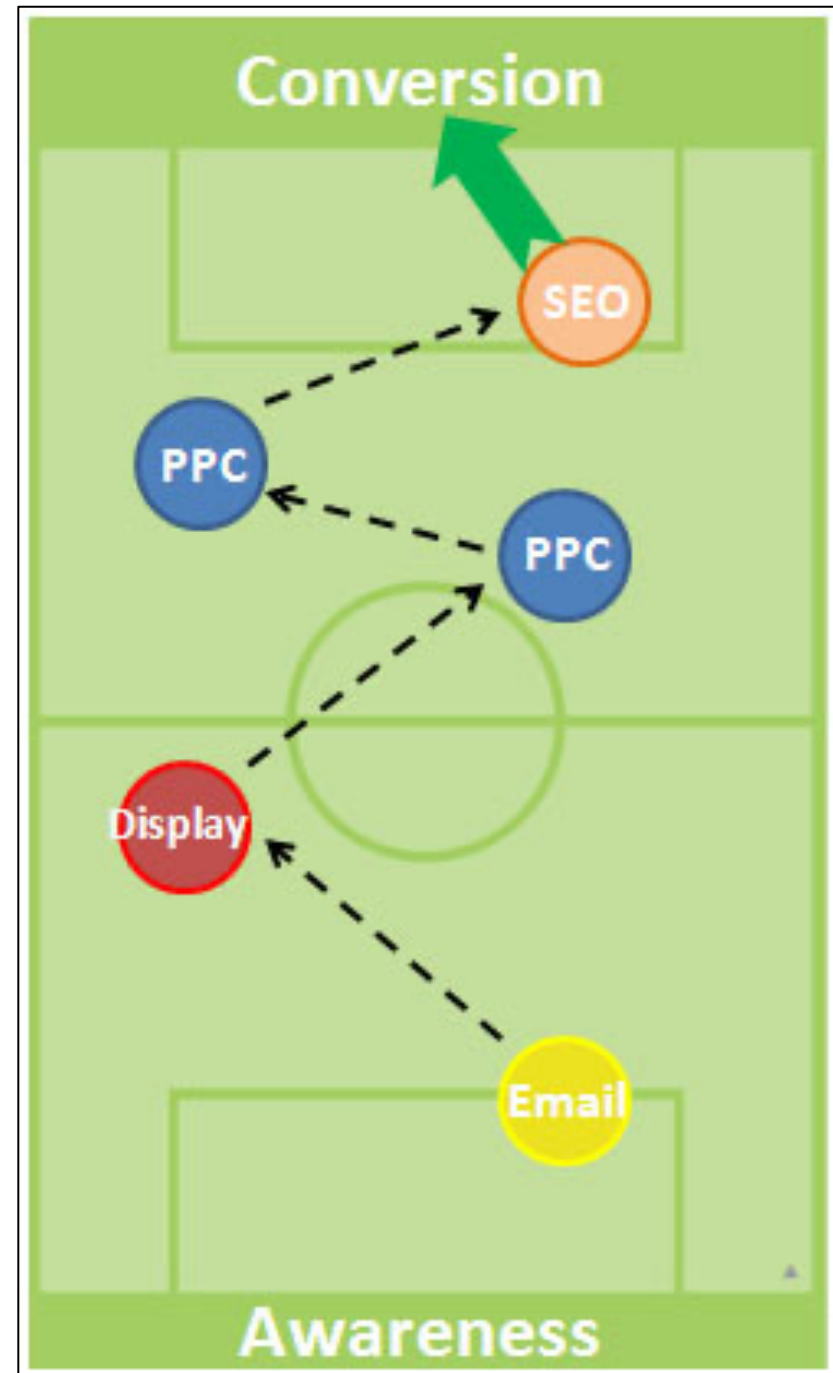


Also known as:

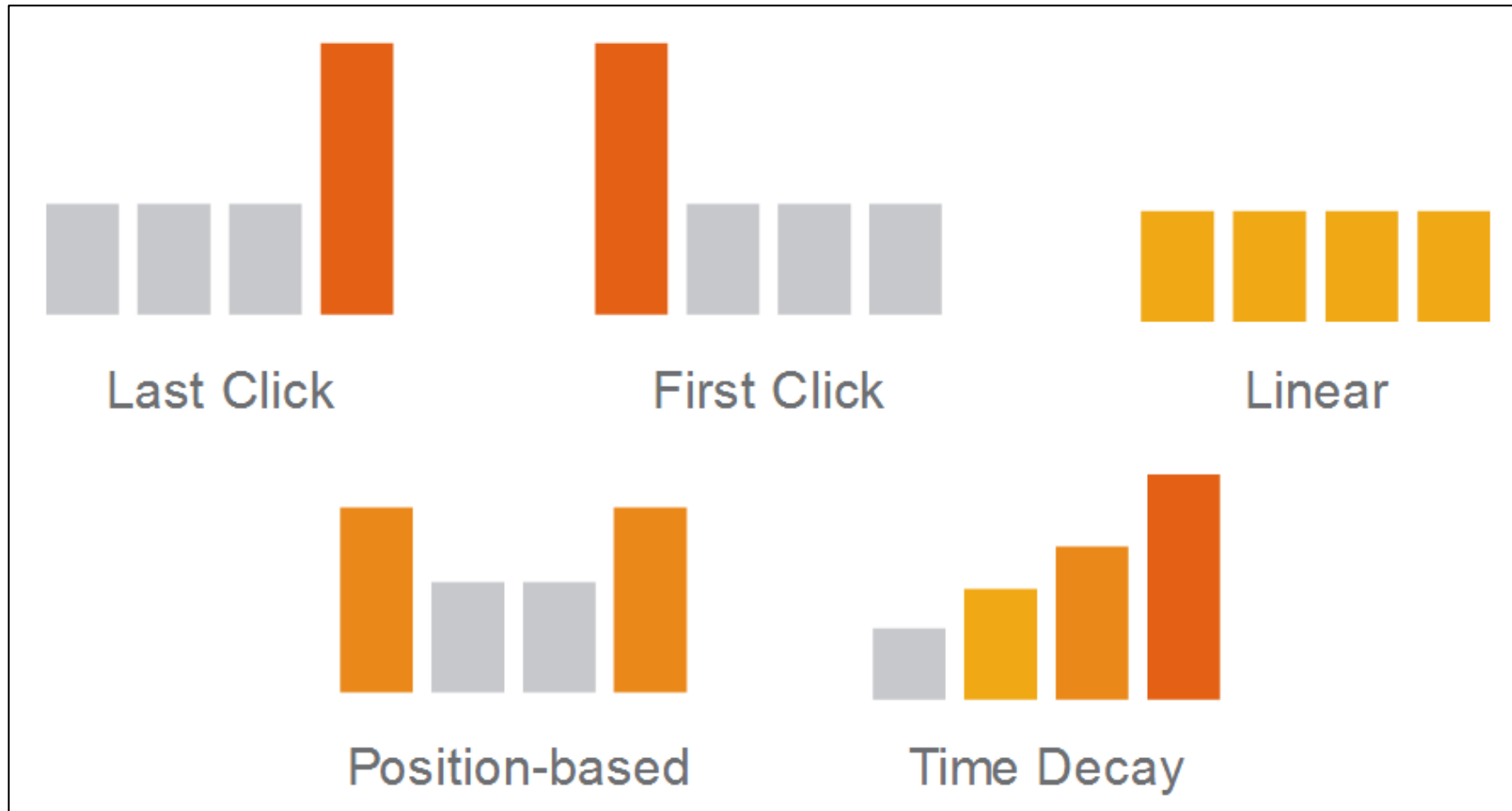
- Multi-channel attribution
- Cross-channel attribution
- Multi-touch attribution
- Attribution Modeling
- ...

Conversion Attribution

Different ads can have different purposes
just like different football players



Rule-Based Attribution Models





Rule-Based Attribution Models

- Most used: “Last Click”
 - ❑ No value for ads that “trigger” interest
 - ❑ Misused by discount or coupon sites
- All previously mentioned models have strengths and weaknesses
- Need for more **data driven** attribution models



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Conversion Attribution: Marginal Contribution

Example



- Marginal contribution of nu.nl advertisement to click-path B:
- $CR_B - CR_A = 0.12\% - 0.1\% = 0.02\%$

Example

A						0.04%
B						0.05%
C						0.08%
D						0.10%
E						0.12%

Attribution for path E

Channel	Marginal Contribution	Attribution
Email	0.04	33%
SEA (Google)	0.01	8%
Direct	0.03	25%
SEA (Google)	0.02	17%
Display (nu.nl)	0.02	17%
	0.12	100%

So, if there are 100 conversions for click-path E with a total value of €5.000, then our banner on nu.nl is awarded a value of $17\% * €5.000 = €835$ (+ the value it added to any other click-paths)

Marginal Contribution

- $Y = 1$ if conversion or 0 if no conversion
- T_{kj} = touch-point k of click-path j
- S_{ij} = set of the first i touch-points of click-path j , $S_{ij} = \{T_{1j}, T_{2j}, \dots, T_{ij}\}$
- $P(Y=1 \mid S_{ij})$: Conversion probability for the Click path S_{ij}
- MC_{ij} = marginal contribution of touch-point i to click-path j

$$MC_{ij} = P(Y = 1 \mid S_{ij}) - P(Y = 1 \mid S_{i-1,j})$$

- Increase in conversion probability caused by touch-point i for click-path j

Example

- Very small example with only 3 marketing channels: A, B, C
- Note that channel A has little added value to the conversions
- Maybe A is a website that shows our ad, but not to the right audience for our product

Click-path	Frequency	Conversions	CR
A	6000	1	0.0002
B	2000	60	0.0300
C	3000	120	0.0400
AB	350	11	0.0314
AC	300	12	0.0400
BA	250	8	0.0320
BC	200	14	0.0700
CA	150	7	0.0467
CB	100	5	0.0500
ABC	10	1	0.1000
ACB	20	3	0.1500
BAC	30	4	0.1333
BCA	40	3	0.0750
CAB	50	7	0.1400
CBA	60	5	0.0833

Example

- Conversion Attribution for path CAB

Click-path	Frequency	Conversions	CR
A	6000	1	0.0002
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BCA	40	3	0.0750
CAB	50	7	0.1400
CBA	60	5	0.0833

Example

Conversion Attribution for path CAB:

- $MC_{C,CAB} =$
 $P(Y = 1 | C) - P(Y = 1 | \emptyset) =$
 $0.04 - 0 = 0.04$
- $MC_{A,CAB} =$
 $P(Y = 1 | CA) - P(Y = 1 | C) =$
 $0.0467 - 0.04 = 0.0067$
- $MC_{B,CAB} =$
 $P(Y = 1 | CAB) - P(Y = 1 | CA) =$
 $0.14 - 0.0467 = 0.0933$

Click-path	Frequency	Conversions	CR
A	6000	1	0.0002
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Example

Conversion Attribution for path CAB:

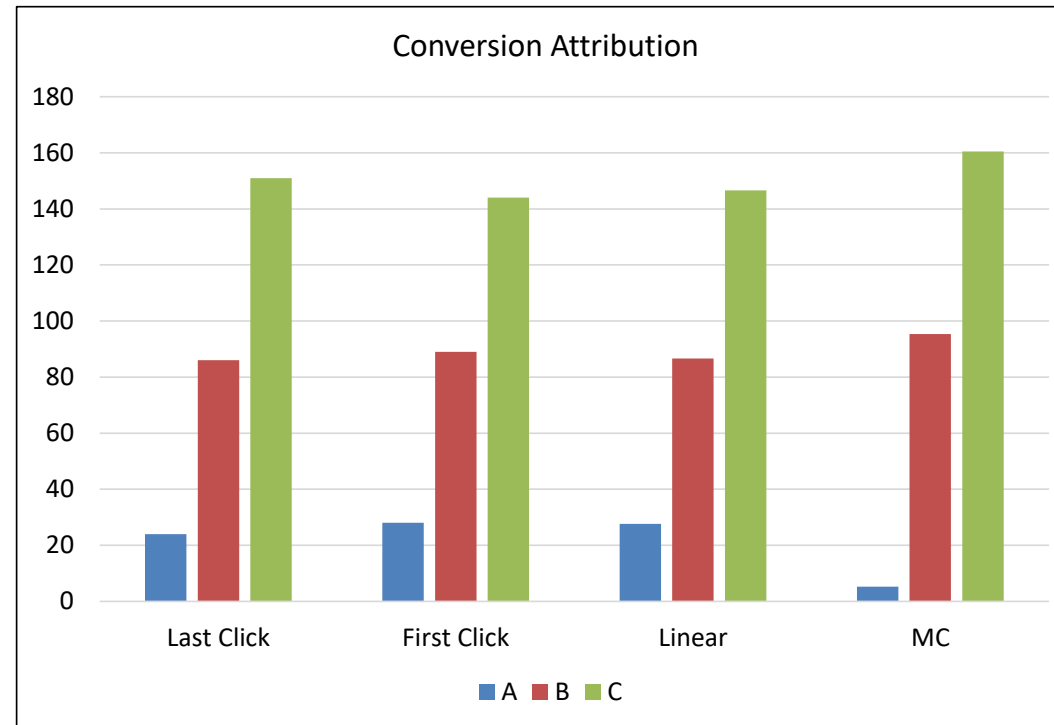
Attributed Conversions for touch-point k
in path j =

$$\text{Conversions}_j * \frac{MC_{k,j}}{\sum_l MC_{l,j}}$$

Touch point	$MC_{k,CAB}$	Attributed Conversions
A	0.0067	$7 * 0.0067 / 0.14 = 0.3$
B	0.0933	$7 * 0.0933 / 0.14 = 4.7$
C	0.0400	$7 * 0.0400 / 0.14 = 2.0$
	0.1400	7

Click-path	Frequency	Conversions	CR
A	6000	1	0.0002
B	2000	60	0.0300
C	3000	120	0.0400
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Example – Overall Conversion Attribution



The Marginal Contribution approach captures the fact that channel A has little value much better than Last Click, First Click or Linear attribution

Methods

- **Marginal Contribution**: based on adding channel i in click-path j , considering specific order of the customer journey j and its subsets. There is only one subset where the channel can be added to in that specific order.

$$MC_{ij} = P(Y = 1|S_{ij}) - P(Y = 1|S_{i-1,j})$$

The Big Challenge

- We need to estimate: $P(Y = 1 | S_{ij})$ for all values of i and j
- But customers can have click-paths of 50+ touchpoints
- Not many customers have the exact same click-path
- This means that each click-path j typically has 0 or 1 observations (usually the very short click-paths do have more observations)
- We can't simply observe $P(Y = 1 | S_{ij})$
- We need to approximate MC_{ij} in a smart way

Possible solutions

- Direct calculation of the Marginal Contribution is generally not possible because we have too few observations to determine $P(Y = 1|S)$
- Dalessandro et al. (2012) disregard the ordering in S to aggregate observations, e.g. $P(Y = 1|ABC) = P(Y = 1|CAB) = P(Y = 1|BCA) = \dots$ So $P(Y = 1|ABC)$ is estimated based on all click paths that contain (only) the touchpoints A, B and C, regardless of their order in the path.
- Shao & Li (2011) go even further and estimate $P(Y = 1|A)$ based on all click-paths that contain A, regardless of the other channels in the path. $P(Y = 1|AB)$ is estimated based on all click-paths that contain both A and B. etc. They don't consider higher order interactions with 3 channels or more (such as ABC) in computing MC.

Estimating the conversion rate

- So far, we used:

$$P(Y = 1) = \frac{N_{Y=1}}{N_{Y=0} + N_{Y=1}}$$

- We simply used the “observed mean” to estimate the conversion rate
- Unreliable results if N is small (which it typically is for many click-paths)
- Let's try to use a regression model to predict $P(Y = 1)$ using all observations



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Conversion Attribution: Logistic Regression

Logistic Regression

$$Y = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots)}}$$

with

- Outcome: $Y = 0$ or 1 (No Conversion or Conversion)
- Features: X = set of variables representing the properties of the customer's click-path
- Each observation is a customer click-path that ends with a conversion or not



Explanatory Variables

- # visits
 - # page views per visit
 - total time on website
 - average time per page
 - time since last visit
-
- All good variables to model conversion probability, but not very useful for Conversion Attribution
 - We need variables that say something about the marketing channels

Dummy per marketing channel

- X_{ij} = dummy whether marketing channel j is in the click-path of customer i
- Example:
 - Assume we have 5 marketing channels: A, B, C, D and E
 - If customer i has click-path ABD, then $X_i = (1, 1, 0, 1, 0)$
 - $P(Y = 1 \mid \text{ABD}) = P(Y \mid X = (1, 1, 0, 1, 0))$
- Notes:
 - $P(Y = 1 \mid \text{ABD}) = P(Y = 1 \mid \text{DAB}) \rightarrow$ we don't consider the ordering of the channels within the click-path
 - $P(Y = 1 \mid \text{AABDBA}) = P(Y = 1 \mid \text{DAB}) \rightarrow$ we don't consider how often a channel appears in the click-path

Example

- 4 marketing channels: A, B, C and D
- Logistic regression with a dummy per channel
- Estimated coefficients given in table

Variable	Coefficient
Constant	-5
A	0.5
B	1
C	0.75
D	1.25

- $P(Y = 1 \mid AB) = \frac{1}{1 + e^{-(-5 + 0.5 + 1)}} = 0.011$
- $P(Y = 1 \mid ABC) = \frac{1}{1 + e^{-(-5 + 0.5 + 1 + 0.75)}} = 0.029$
- Marginal Contribution of C to ABC is 0.018

Frequency & Touch-Point Order

- Until now in the Logistic Regression we only considered if a channel is in a click-path or not (just like Shao & Li (2011))
- We did not consider:
 - The ordering of the channels within the click-path
 - How often a channel appears in the click-path
- We can do this by using the following variables:
 - X_{ijk} = dummy whether marketing channel j is the k^{th} touch-point in the click-path of customer i
- But this results in a large number of dummy variables. We will need an estimation technique such as SGD (see slides on CTR modelling)

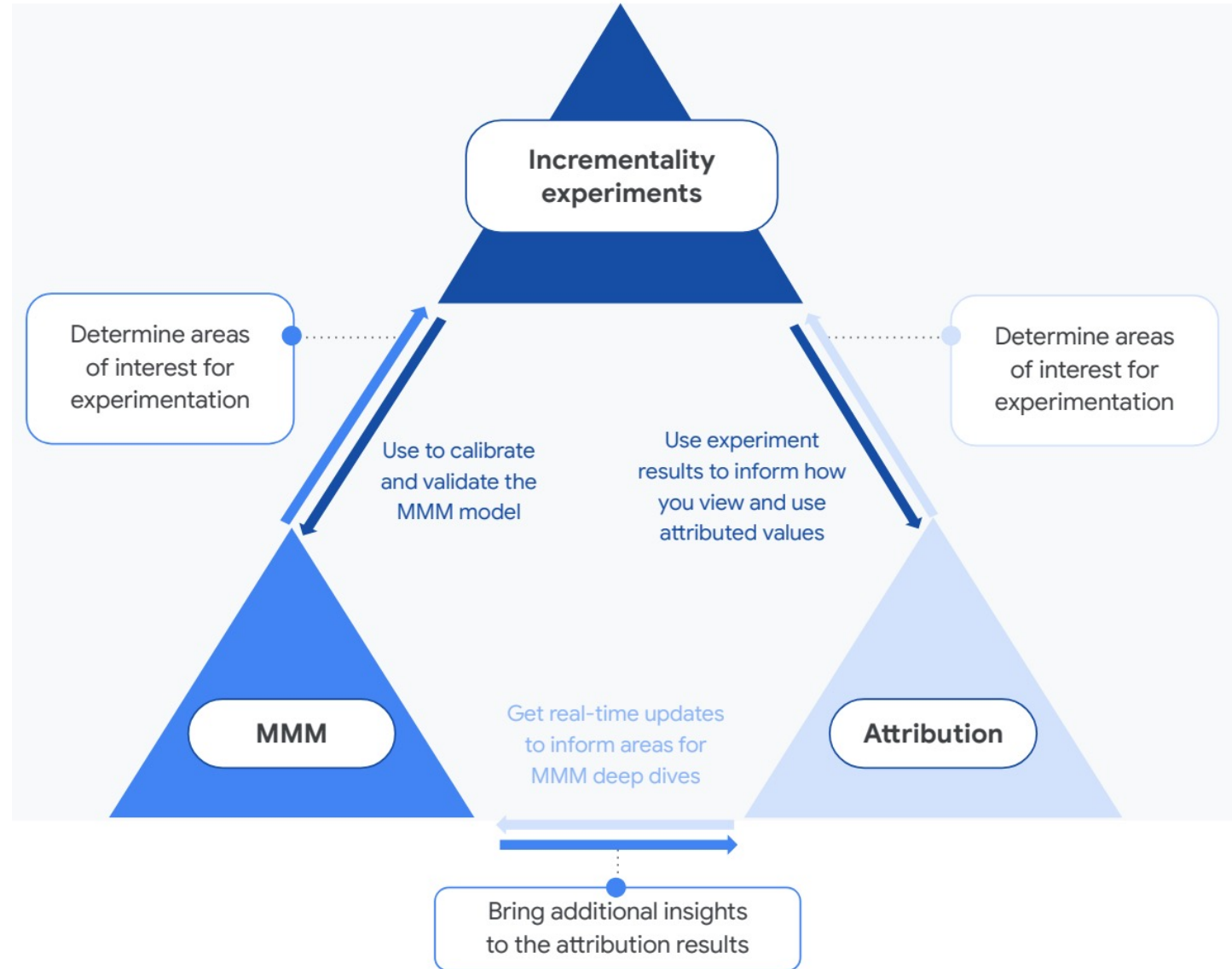
Conversion Attribution

- Relevant topic for online advertisers
- Large data volumes (but few observations per click-path)
- Many companies offer solutions (Adobe, Google, etc.)
- Some known data challenges (see next slide)

Data Challenges

- Privacy: (3rd party) cookie restrictions
- Many click paths not complete because of use of multiple devices and apps
 - Fewer observations of longer paths
 - Companies are looking for smart ways to link your devices
- Noise:
 - Bounces: people that click on an ad accidentally and directly leave within a few seconds
 - Same person seeing the same ad many times within a short time span (i.e. banner on home page of popular news site)
- Not all channels are covered, e.g., offline channels

How to combine?





Today's Course Material

- Lecture notes
- References are included for background information only. They don't need to be studied for the exam.

References

- Dalessandro, Perlich, Stitelman & Provost (2012), Causally Motivated Attribution For Online Advertising, ADKDD'12 Proceedings of the Sixth International Workshop on Data Mining for Online Advertising and Internet Economy
- Shao & Li (2011), Data-driven Multi-touch Attribution Models, Marketing Science, Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 258-264